Corporate Finance

Topic 4 – B

Bernardo Ricca

The asset substitution problem

- Consider a firm that generates a random liquidating payoff of X tomorrow (at $t = 1$)
  - We can think of X as the resale value of the firm
- The firm has debt with a face value of K at $t = 1$
- Debt is risky, meaning that with positive probability $X \leq K$
- The equity payoff is $\max\{X - K, 0\}$, the same payoff of a long position in a call option on the assets of the firm, X, with strike K
  - The equityholder bears the upside (any quantity above K accrues to him/her), and obtains zero whenever losses are such that $X < K \Rightarrow$ preference for **riskier** projects

The asset substitution problem

- The debt payoff is $\min\{X, K\} = K - \max\{0, K - X\}$ the same payoff of a long position in a risk-free bond with face value K and a short position in a put option on the assets of the firm, X, with strike K
  - Debtholders have priority and they only care about the first K dollars $\Rightarrow$ preference for **safer** projects
- Different payoffs $\Rightarrow$ conflict of interest between equity and debtholders
- The preferred choices of debtholders and shareholders can differ and are not necessarily equal to the choice that maximizes firm value
- As equityholders have control rights, firms tend to pick riskier projects when leverage is high, and not necessarily the ones that maximize firm value (the asset substitution or risk-shifting problem)

Asset substitution: an example

- Everyone is risk-neutral ($\Rightarrow$ discount rates are always equal to the risk-free rate)
- The risk-free rate is $r_f = 20\%$
- The assets of the firm today ($t = 0$) consist of a cash balance of 100
- The firm contracted debt with face value 120 in the past that is due at $t = 1$
- Three states at $t = 1$: bust (with probability 20%), neutral (60%) and boom (20%)

Asset substitution: an example

- Three investment opportunities
- Project 1 is the one that maximizes the firm value

Table: Projects payoffs

		Payoff (t=1)				
	Investment (t=0)	Bust	Neutral	Boom	Exp. payoff	NPV
Project 1	-100	110	130	150	130	8.3
Project 2	-100	60	90	240	114	-5.0
Project 3	-100	120	122	125	122.2	1.8

Asset substitution: an example

- Yet, equityholders prefer project 2...

Table: Equityholders' payoffs

	Payoff (t=1)				
	Bust	Neutral	Boom	Exp. payoff	Market value
Project 1	0	10	30	12	10
Project 2	0	0	120	24	20
Project 3	0	2	5	2.2	1.8

Asset substitution: an example

- ... whereas debtholders prefer project 3

Table: Debtholders' payoffs

	Payoff (t=1)				
	Bust	Neutral	Boom	Exp. payoff	Market value
Project 1	110	120	120	118	98.3
Project 2	60	90	120	90	75.0
Project 3	120	120	120	120	100

Asset substitution: an example

- Shareholders have control rights and thus project 2 is chosen
- The project that maximizes firm value is not chosen: the presence of risky debt leads to inefficiency
- The (market value) balance sheet of the firm at $t = 0$ is $A = 95$, $D = 75$ and $E = 20$

Asset substitution: an example

- Suppose now the firm has no cash at $t = 0$ and needs to raise 100 from outside investors
- Suppose the firm attempts to raise 100 by issuing debt
- Are debtholders willing to price the bond as a risk-free asset and ask for only $100(1 + r_f) = 100(1 + 20\%) = 120$ at $t = 1$?

Asset substitution: an example

- Suppose now the firm has no cash at $t = 0$ and needs to raise 100 from outside investors
- Suppose the firm attempts to raise 100 by issuing debt
- Are debtholders willing to price the bond as a risk-free asset and ask for only $100(1 + r_f) = 100(1 + 20\%) = 120$ at $t = 1$?
- No! The risk-shifting problem implies that equityholders would undertake project 2
- Debtholders would suffer a loss: they would "pay" 100 for an asset that is worth 75

Asset substitution: an example

- Debtholders anticipate the problem and do not provide finance (market breakdown) in this case because the preferred choice, project 2, has a negative NPV
  - As the project destroys value, it is impossible for lenders to get back their investment of 100
- The risk-shifting problem prevents the financing of a firm that has profitable investment opportunities
- What if the preferred choice has a positive NPV?

Asset substitution: an example

- If the preferred choice has a positive NPV, financing would take place but at a higher interest rate

- Consider the following alternative payoffs for project 2:

Table: Alternative project payoffs

		Payoff (t=1)				
	Investment (t=0)	Bust	Neutral	Boom	Exp. payoff	NPV
Project 1	-100	110	130	150	130	8.3
Project 2	-100	80	100	240	124	3.3
Project 3	-100	120	122	125	122.2	1.8

- Anticipating the choice of project 2, lenders would charge an interest rate of 120% (a face value of 220)!

- At this interest rate the market value of debt is 100

Asset substitution: an example

- What about outside equity financing? Let's return to the original values of the example
- If the firm raises 100 by selling a fraction α to outside investors, original owners retain a share 1 – α of the firm
- No risk-shifting problem: they choose the project that maximizes the firm value: project 1

	Firm value	Inside equity	Outside equity
Project 1	108.3	(1-α)108.3	α 108.3
Project 2	95	(1-α)95	α 95
Project 3	101.8	(1-α)101.8	α 101.8

- There is no conflict of interest between inside and outside equity (same claim)

Asset substitution: an example

- Given that project 1 is chosen, outside investors invest 100 in exchange for a share

$$\alpha = \frac{100}{108.3}$$

of the firm

- Financing takes place and the choice that maximizes surplus is chosen

Asset substitution: implications

- Firms whose business activities are such that managers (acting in the interest of shareholders) have more discretion with regard to risk choices should have less debt and prioritize equity financing
- On the other hand, firms in which managers have little discretion with regard to risk can have more debt
  - E.g. mature industries, regulated public utilities
- Risk-shifting distortions are higher when the firm is in financial distress (gambling for resurrection)
  - In particular, owners might delay filing for bankruptcy so as to keep the option value of equity alive

Asset substitution: mitigation

- Our framework is static (one-period)
- In a multi-period setting the need to access markets in the future generates an incentive to build a good reputation and can mitigate the problem
- Jensen and Meckling (1976):

  It seems clear, for instance, that the expectation of future sales of outside equity and debt will change the costs and benefits facing the manager in making decisions which benefit himself at the (short–run) expense of the current bondholders and stockholders. If he develops a reputation for such dealings, he can expect this to unfavorably influence the terms at which he can obtain future capital from outside sources. This will tend to increase the benefits associated with "sainthood" and will tend to reduce the size of the agency costs.

Asset substitution: mitigation

- Short-term debt: frequent repricing that incorporates any increases in risk and the threat to not be able to rollover the debt lessen incentives for excessive risk taking

- Covenants (provisions that restrict the actions that firms can take): e.g. prohibit investments in new, unrelated lines of business; indirect effects of strict covenants if safer projects enable firms to meet the required ratios (e.g. safer projects generate cash flows earlier, as in Gilje, 2016)

- Convertible debt: debtholders can exercise the option to become equityholders if the business booms

  - Equityholders are no longer the sole residual claimants in case of success: part of the gains accrue to debtholders

Asset substitution: empirical evidence

- Very challenging: exogenous variation in leverage + measurement of the riskness of firm investment
- Gilje (2016): firms *reduce* risk when they approach distress
  - Sample: firms in the oil & gas industry
  - Evidence for mitigating factors such as short-term debt and covenants

Asset substitution: friction

- What is the friction underlying the risk-shifting problem?
- Commitment: managers (acting on behalf of shareholders) cannot commit to taking good investments once debt is contracted
- Similar to the Holmstrom and Tirole (1997) model, shareholders pay the cost by not receiving funding or paying a higher interest rate
- If equity is costly/difficult to raise, firms might forgo profitable investment opportunities
- Impossible to write contracts on project choices to guarantee such a commitment

Debt overhang

- After debt is contracted, profitable investments connected to the assets in place of a firm might not be undertaken
- The reason is that borrowers (existing shareholders) bear all the investment costs (money, time, effort) but part of the returns accrue to existing creditors
- In other words, debt in place acts as an implicit tax on investment, the tax rate being related to the probability of default

Debt overhang (cont'd)

- The problem is more severe when default is more likely, as the portion of returns that accrue to lenders (the 'tax burden') increases
- The problem is less severe if the value of the relationship with the original creditors is large or if default costs are increasing in the amount owed (dynamic repayment incentives)
- Originally a conjecture about firms' behavior (Myers, 1977), the notion of debt overhang has been extended (and documented) to individuals

Some empirical evidence

- Borrowers with negative home equity (value of the house is smaller that the mortgage balance) invests less in home improvements and maintenance (Melzer, 2017)

- Borrowers that are heavily indebted will share a significant part of their future income with creditors, potentially affecting their labor supply decisions (time, effort, search, occupation). See Dobbie and Song (2020) and Di Maggio et al. (2020)

- We say that debt introduces ex post inefficiency:

  - The efficiency loss is larger for investment opportunities whose value added wane if not undertaken immediately

  - Example: investments in education, house maintenance investments such as fixing a foundation problem or a plumbing leak

An example

- For simplicity, the risk-free rate is zero, existing debtholders and equityholders are risk-neutral, and the firm is owned by a single equityholder
  - Risk-neutrality means that the risk premium is zero, so discount rates are $r = r_f + \text{risk-premium} = r_f$, where r_f is the risk-free rate. Since the $r_f = 0$, cash flows at $t = 1$ are discounted by $1 + r = 1$
- The firm has assets in place A that have a (market) value at date 1 of
  - 500 with probability 0.5 (*boom state*)
  - 200 with probability 0.5 (*slump state*)
- The firm has outstanding debt with face value $F = 300$ due at date 1

An example (cont'd)

- As debt is senior to equity, the current market values (at date 0) are

Table: Baseline market values of A, D, and E

	Boom	slump	Market value
Assets (A)	500	200	350
Debt (D)	300	200	250
Equity (E)	200	0	100
$A = D + E$	500	200	350

- Remark: note that the market value of debt is smaller than its face value even with a zero discount rate (250<300). The bond is selling at a discount due to a positive default probability

An example (cont'd)

- Suppose the firm owner has access to project that requires an investment of 60 at date 0 and increases the value of the assets in place by 100 at date 1 in both states (boom and slump)
- The project is profitable: it has a positive NPV of $100 - 60 = 40 > 0$. Therefore the investment should be undertaken in a first-best allocation
- Assumption: the cash flows of the project cannot be obtained separately from those of the assets in place (the project needs to be carried out "inside" the firm). Otherwise the project can be undertaken in separate entity with no existing debtholders

An example (cont'd)

- Given that the project is profitable, is the firm owner willing to invest in this project?
- The answer is no!
- If the firm owner decides to spend 60 and undertake this project, the market values would change to

Table: Market values of A, D, and E if the project is undertaken

	Boom	slump	Market value
Assets (A)	600	300	450
Debt (D)	300	300	300
Equity (E)	300	0	150
$A = D + E$	600	300	450

An example (cont'd)

- Even though the project has a positive NPV of 100-60=40, if the owner invests 60, the equity value increases only by 50, from 100 to 150
- The reason is that the firm owner bears the entire investment cost but part of the gain goes to debtholders: the market value of debt increases from 250 to 300
- This is the debt overhang problem: profitable investment opportunities might not be pursued if the borrower is heavily indebted
- As the investment is not undertaken, the market values remain $D = 250$ and $E = 100$

An example (cont'd)

- **Renegotiation** can be a win-win solution
- Suppose the firm and the debtholders agree on a new face value of debt, say $F' = 280 < 300$, conditional on the entrepreneur undertaking the project (this condition can written down in a contract)
- If the firm owner decides to undertake this project, the market values would change to

Table: Market values of A, D, and E if the project is undertaken and debt is renegotiated

	Boom	slump	Market value
Assets (A)	600	300	450
Debt (D)	280	280	280
Equity (E)	320	20	170
$D + E$	600	300	450

An example (cont'd)

- Now the increase in equity value is $70 = 170 - 100$, which is bigger than the investment cost, 60
- The equityholder is willing to make the investment, as

$$\underbrace{170}_{\text{new equity value}} - \underbrace{60}_{\text{investment}} = 110 > \underbrace{100}_{\text{previous equity value}}$$

- The market value of debt increases from 250 to 280
- With renegotiation, both parties are better off

An example (cont'd)

- We assumed that renegotiation takes place conditional on the entrepreneur undertaking the project
- What if debtholders cannot force the entrepreneur to invest (unconditional renegotiation)?
- If the entrepreneur renegotiates and does not invest the payoffs are

Table: Market values of A, D, and E if the project is not undertaken and debt is renegotiated

	Boom	slump	Market value
Assets (A)	500	200	350
Debt (D)	280	200	240
Equity (E)	220	0	110
A=D+E	500	200	350

An example (cont'd)

- The payoffs with renegotiation are the same with (170-60=110) or without the investment (110), thus we can assume that the investment is undertaken even in an unconditional renegotiation
- Remark: the new face value of 280 is of the of the possible outcomes of the renegotiation
  - In general, the new face value will depend on the bargaining power of the parties

Debt overhang: mitigation

- Ex post ways: issuance of senior claims to existing debt (issuance of claims junior to existing debt would not solve the problem)
  - Sometimes not feasible because of covenants
  - Rationale for debtor-in-possession (DIP) financing: lending to a firm that filed for bankruptcy protection under Chapter 11
  - This type of debt is senior to existing claims and allows firms in distress to keep investing
- Ex ante ways: debt structure that is easier to renegotiate, such as banks rather than public debt (bonds) and fewer creditors

The benefits of debt renegotiation

- Debt renegotiation alleviates the debt overhang problem and can restore (ex post) incentives
  - Investment decisions are less distorted
  - Both creditors and debtors are better off (win-win)
  - Smaller face value, yet smaller probability of default or smaller loss given default
- In economic jargon, we say that renegotiation can be ex post efficient

The benefits of debt renegotiation (cont'd)

- If the borrower is risk-averse, renegotiation also provides insurance to "bad luck" shocks, such as unemployment in a recession, health shocks, etc.
- By keeping the borrower with a clean credit record and a manageable debt payment plan, renegotiation might allow the borrower to retain some access to credit markets
- Individuals that are less liquidity constrained have more time to search for better jobs and negative credit information can adversely affect labor market prospects (Bos et al., 2018)
- Individuals might be able to keep physical assets that are important to sustain their labor income (e.g. motorcycle) or the enrollment in an education program

The benefits of debt renegotiation (cont'd)

- A wave of defaults (and liquidations) in a recession can amplify shocks to the banking sector
- Defaults also have externalities:
  - Example: foreclosed houses deplete the prices of non-foreclosed houses in the same neighborhood (Campbell et al., 2011)

The prevalence of private renegotiation

- Out of court loan renegotiation is common
  - In the US, even though individuals can choose to file for bankruptcy under Chapters 7 or 13, the majority of debt restructuring take place outside the bankruptcy system (Dawsey et al., 2013)
  - According to the Central Bank of Brazil, 4.6 million borrowers were overindebted in December 2019 (more than 90 days overdue, ratio debt service to income larger than 50%) and around 252 thousand individuals per month restructure their debts[1]
- Private renegotiation is potentially more relevant in places with inefficient judicial systems, where it is too costly and ineffective to file a case against a borrower in default

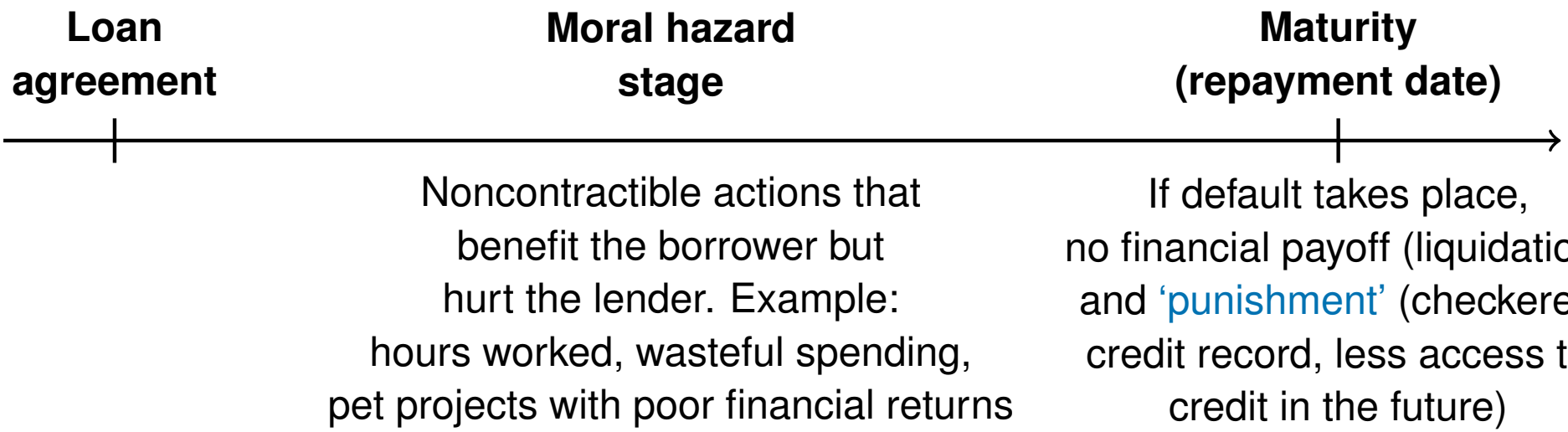
[1] *Relatório de Economia Bancária*, 2018 and 2019.

Private renegotiation and public policy

- Renegotiation can be influenced by policy
  - Accounting and regulatory (capital requirements, loan loss provision) treatment of restructured operations
  - Personal bankruptcy law / institutional environment
- Should governments promote renegotiation (or forbearance)? Examples:
  - Coronavirus Aid, Relief, and Economic Security (CARES) Act during the COVID-19 pandemic
  - Home Affordable Modification Plan (HAMP) during the Great Recession
  - Ongoing debate about cancelling student-loan debt

The costs of debt renegotiation

- As we saw, under certain assumptions, the debt contract arises as an optimal contract in the presence of moral hazard, in the sense that it is the best way to provide correct incentives



The costs of debt renegotiation (cont'd)

- The **prospect of renegotiation** makes the debt contract **soft**
- The anticipation of renegotiation in the future undermines incentives at the moral hazard stage



Noncontractible actions that benefit the borrower but hurt the lender. Example: hours worked, wasteful spending, pet projects with poor financial returns

If default takes place, borrowers have some financial payoff and face a less harsh 'punishment' (clean credit record, drop in access to credit is not so severe)

The costs of debt renegotiation (cont'd)

- By undermining incentives, **the prospect renegotiation in the future** can increase the default risk of debt contracts, leading to credit rationing and higher costs
- The prospect of renegotiation can also trigger a strategic default: individuals can claim they do not have enough cash to repay the loan even if they have so, in order to obtain better conditions (Mayer et al., 2014)
  - Strategic default differs from a liquidity default, a situation in which the borrower is not able to repay because he does have enough cash

The costs of debt renegotiation (cont'd)

- In economic jargon, we say that the prospect of renegotiation can be ex ante inefficient
- The moral hazard costs of renegotiation are similar to the concern that too-big-to-fail banks take too much risk because they know that they will be bailed out in case of insolvency
- Public policy should take into account costs and benefits

Default as insurance

- Based on Indarte (2023)
- In the next period, an individual has
  - A debt payment D
  - Liquid assets $A < D$, which lenders cannot seize
  - Income $Y \in [\underline{Y}, \bar{Y}]$ distributed according to a cdf $F(.)$

$$Pr[Y \leq a] = F(a)$$

- The individual is risk-averse with utility $u(C)$, where $u(0) = 0$, $u' > 0$ and $u'' < 0$

Default as insurance

- In case of default, courts determine that the individual retains minimum income level E, where $E < \underline{Y}$
  - Creditors receive $Y - E$ (wage garnishment) up to the amount due D
- Default entails a non-pecuniary cost $\sigma > 0$
- If $A + Y < D$, individuals cannot repay (cash flow default)
- If $A + Y \geq D$, individuals can choose not to repay (strategic default)
  - If they repay, they consume $A + Y - D$
  - If they default, they consume $A + E$

Default as insurance

- When $A + Y \geq D$ (or $Y \geq D - A$), the individual defaults whenever

$$u(A + Y - D) < u(A + E) - \sigma$$

- As long as $u(A + E) - \sigma > 0$, there is a $Y^* > D - A$ such that

$$u(A + Y^* - D) = u(A + E) - \sigma$$

- The individual defaults whenever $Y < Y^*$
  - Default acts as insurance whenever the income drops below a given level $\Rightarrow$ it guarantees a consumption of $A + E$ when income is too low
  - Since the individual is risk-averse, default can increase welfare by making the individual bear less income risk

Default as insurance

- The default probability is then $p = Pr[Y \leq Y^*] = F(Y^*)$
- An increase in the generosity of debt relief E provides more insurance...

  ... but it also increases default rates (and thus interest rates in equilibrium)

$$\frac{\partial p}{\partial E} = F'(Y^*)\frac{u'(A+E)}{u'(A+Y^*-D)} > 0$$

Identifying information asymmetries in credit markets

- In general, it is difficult to disentangle adverse selection and moral hazard
- Consider the following experiment:
  - Randomly allocate individuals into different groups
  - Then offer the same loan contract at different prices (interest rates) to the groups of otherwise identical individuals
- A positive correlation between interest rates and default can arise due to
  - Adverse selection: those with a relatively high probability of default are more likely to accept a high rate; or
  - Moral hazard: a high interest rate can undermine incentives and increase the default probability

Karlan and Zinman (2009): research design

- Randomization of interest rates across three dimensions:
  - Interest rate offered in a direct-mail solicitation: *the offer rate*
  - Actual interest rate on the loan contract: *the contract rate*
  - Interest rate on future loans (conditional on not defaulting): *dynamic repayment incentive*
- The design enables the comparison of
  - Borrowers who select in at different rates and then face identical repayment incentives: adverse selection
  - Borrowers who select at identical rates and then face different repayment incentives: moral hazard

- Possible to disentangle:
  - Hidden information, defined as ex-ante selection effects, which includes both selection on unobserved risk and selection on unobserved anticipated effort

  from
  - Hidden action: ex-post incentive effects

- The design cannot disentangle the two sources of hidden information

Model

Three stages:

- **Stage 1:** the individual decides whether or not to take up a loan with an exogenous interest rate r^o. The rate r^o does not convey any information about interest rates of future loans: the individuals believe that future loans will be offered at a normal rate r.

- **Stage 2:** The lender randomly lowers the interest rate for some borrowers to $r^c < r^0$. With an independent randomization, the lender lowers the interest rate of future loans to $r^f = r^c < r$ for some borrowers (conditional on repaying the loan). Given r^c and r^f the borrower chooses how much effort to put into generating cash flows to repay the current loan

- **Stage 3:** Cash flows are realized and the borrower decides whether or not to repay the loan

- A penniless risk-neutral individual has access to a project that requires an investment of $I = 1$
- The loan can be obtained from a single lender
- There is no need to model the lender's behavior: the decisions are randomized
- Individuals are indexed by a risk type $\theta_i \in [\underline{\theta}, \bar{\theta}]$ and exert effort $e \in [\underline{e}, \bar{e}]$ at a cost $c(\theta_i, e) = e$
- If the project succeeds (state g), it generates (unverifiable) cashflow $Y(\theta_i)$; if it fails (state b), the cashflow is zero
- The probability of success is $\pi(\theta_i, e)$ is a function of both the risk type and the effort provision

- The borrowers is subject to a state-specific cost of default: $C_i(r^f)$, $i = g, b$, where $C_i(r^f)$ is decreasing in r^f and $C_b(r^f) < C_g(r^f)$
  - We can think of $C_i(r^f)$ as the forgone discounted profits from future loans. Hence the assumption that $C_i(r^f)$ is decreasing in r^f
- Remark: The model features both hidden information and hidden action
  - Hidden information manifests itself in the form of unobservable risk type θ_i and anticipated effort

Assumptions:

- **(A1)** $Y(\theta_i) > 1 + r^o$ for all θ_i: if the project succeeds the loan can be repaid at the offer interest rate
- **(A2)** $\frac{\partial \pi(\theta_i,e)}{\partial e} > 0$ and $\frac{\partial^2 \pi(\theta_i,e)}{\partial e^2} < 0$: the probability of success is increasing and concave in effort
- **(A3)** $\frac{\partial \pi(\theta_i,e)}{\partial \theta_i} < 0$: the higher the risk type, the lower the probability of success (and thus the higher the probability of a liquidity default)
- **(A4)** $\pi(\theta_i, e)Y(\theta_i) = \bar{Y}(e)$ for all θ_i: for a fixed e, all risk types generate the same expected return. Projects with a higher θ_i are riskier in terms of second-order stochastic dominance (mean-preserving spread)
- **(A5)** $C_g(r^f) \geq 1 + r^o$ for all relevant r^f: the borrower repays the loan when the project succeeds (there is no strategic default)

- Stage 3: given (A1) and (A5), the borrower repays if and only if the project succeeds

- Stage 2: The borrower with type θ_i solves

$$\max_{e \in [\underline{e}, \bar{e}]} \pi(\theta_i, e)\Big[Y(\theta_i) - (1 + r^c)\Big] + (1 - \pi(\theta_i, e))\Big[- C_b(r^f)\Big] - e$$

- FOC: $\frac{\partial \pi(\theta_i, e)}{\partial e}\Big[Y(\theta_i) - (1 + r^c) + C_b(r^f)\Big] = 1$

- **Hidden action effect 1:** Given (A2), effort is decreasing in r^c

- **Hidden action effect 2:** Given (A1), (A2) and (A5), effort increases as the cost of default increases. Thus, repayment increases when r^f decreases (dynamic repayment incentive)

- Given (A4), $\frac{\partial \pi(\theta_i, e)}{\partial e} Y(\theta_i) = \bar{Y}'(e)$, we can rewrite the FOC as

$$\frac{[1 - \bar{Y}'(e)] Y(\theta_i)}{\bar{Y}'(e)} = C_b(r^f) - (1 + r^c)$$

  Given (A2), $\bar{Y}'(e) > 0$ and $\bar{Y}''(e) < 0$

- Then the optimal effort level $\hat{e}(r^c, C_b(r^f), \theta_i)$ is a decreasing function of the risk type θ_i

- Stage 1: individuals decide to take up a loan with a rate r^o if the expected gains from the project are bigger than the outside option, which yields a zero payoff by assumption.

- Given the optimal effort level $\hat{e}(r^o, C_b(r^f), \theta_i)$ and the belief that future interest rates will be offered at a normal rate r, an individual accepts the offer r^o if and only if

$$\pi(\theta_i, \hat{e}(r^o, C_b(r), \theta_i))\Big[Y(\theta_i) - (1 + r^o) + C_b(r)\Big] - C_b(r) - \hat{e}(r^o, C_b(r), \theta_i) \geq 0$$

Given (A4) we can rewrite

$$\bar{Y}(\hat{e}(r^o, C_b(r), \theta_i)) + \pi(\theta_i, \hat{e}(r^o, C_b(r), \theta_i))\Big[C_b(r) - (1 + r^o)\Big] - C_b(r) - \hat{e}(r^o, C_b(r), \theta_i) \geq 0$$

- If $C_b(r) < 1 + r^o$, then the left-hand side is increasing in θ_i
- Therefore there exists a threshold $\underline{\theta}(r^o)$ such that risk types $\theta_i > \underline{\theta}(r^o)$ take up the loan
- By the implicit function theorem and the envelope theorem

$$\frac{d\underline{\theta}(r^o)}{dr^o} = -\frac{-\pi(\theta_i, \hat{e}(r^o, C_b(r), \theta_i))}{\frac{\partial \pi(\theta_i, \hat{e}(r^o, C_b(r), \theta_i)}{\partial \theta_i}\Big[C_b(r) - (1 + r^o)\Big]} > 0$$

- **Hidden information effect:** A higher offer interest rate leads to a riskier pool of clients and thus a higher default rate
- Note: for a discrete change in r^o, the envelope theorem does not hold: besides the direct effect of r^o, we also have to consider the indirect effect of r^o on the optimal effort choice

Experiment

- Offer rate can be low or high, $r^o \in \{r_L, r_H\}$, with $r_L < r_H$
- Of those that take up the high $r^o = r_H$, some are randomly surprised with a new lower contract rate $r^c = r_L \neq r^o$, while the remainder continue to receive the high rate $r^o = r^c = r_H$
- Hidden information: comparison of the repayment behavior of those that were offered and accepted the low rate, $r^o = r^c = r_L$, with those that were offered and accepted the high rate, $r^o = r_H$, but ended up with a low contract rate $r^c = r_L$
- Hidden action: comparison of the repayment behavior of those that took up the high rate and were not offered a low contract rate, $r^o = r^c = r_H$, with those that were offered and accepted the high rate, $r^o = r_H$, but ended up with a low contract rate $r^c = r_L$

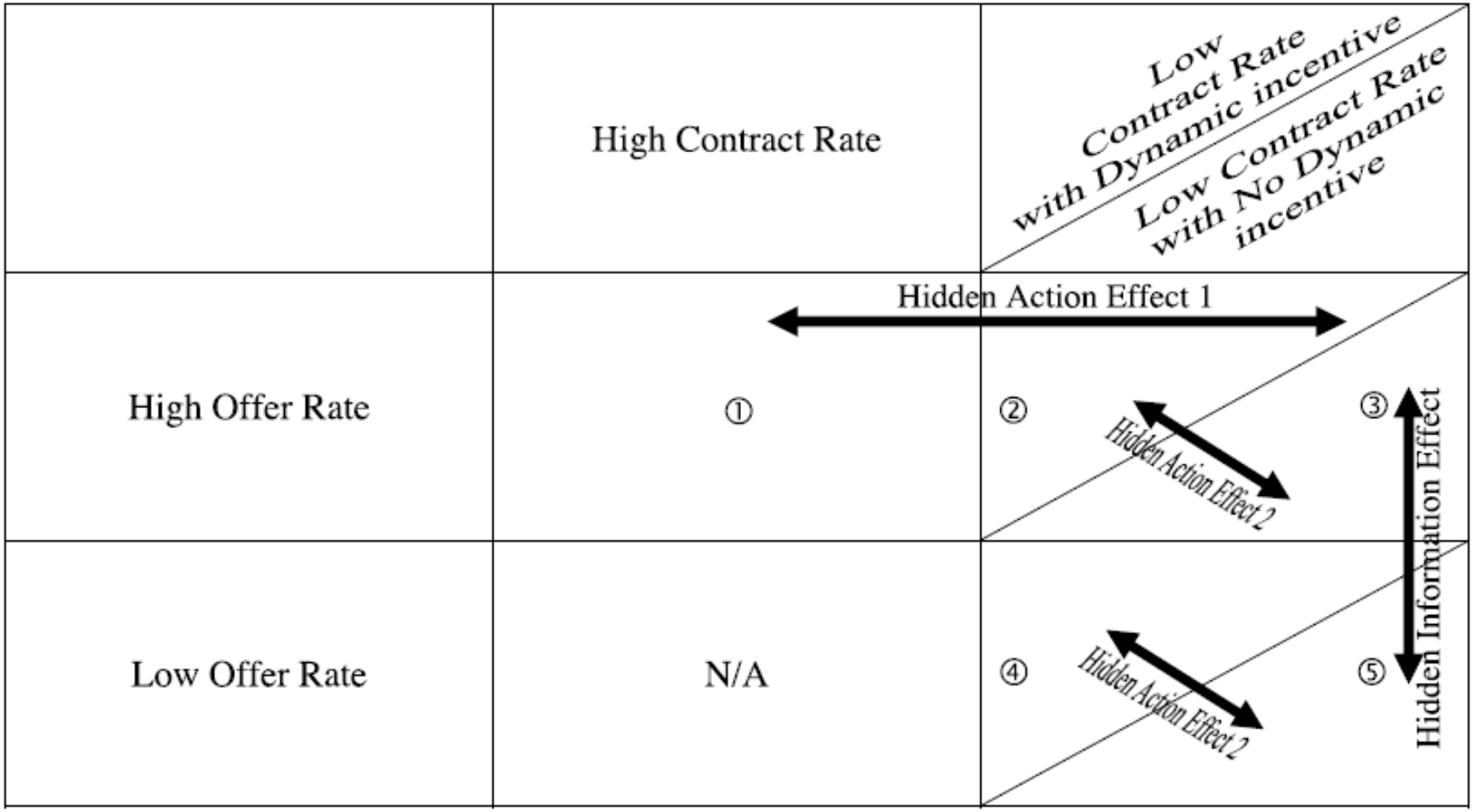


FIGURE 1.—Some basic intuition for our identification strategy.

Regression

- Empirical model

$$Y_i = \alpha + \beta_o r^o + \beta_c r^c + \beta_b C + X_i + \epsilon_o$$

  where Y_i is a measure of default

- $\beta_c > 0$ identifies any *hidden action effect 1* (effect of a higher repayment on effort)
- $\beta_b < 0$ identifies any *hidden action effect 2* (effect of higher dynamic incentives on effort)
- $\beta_o > 0$ identifies adverse selection (*hidden information effect*, effect of a higher offer rate on default)

TABLE I

EMPIRICAL TESTS OF HIDDEN INFORMATION AND HIDDEN ACTION: FULL SAMPLE

	OLS							
Dependent Variable:	*Monthly Average Proportion Past Due*		*Proportion of Months in Arrears*		*Account in Collection Status*		*Standardized Index of Three Default Measures*	
Mean of Dependent Variable:	0.09	0.09	0.22	0.22	0.12	0.12	0	0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Contract rate (Hidden Action Effect 1)	0.005	0.002	0.006*	0.002	0.001	−0.001	0.014	0.004
	(0.003)	(0.004)	(0.003)	(0.004)	(0.005)	(0.005)	(0.011)	(0.013)
Dynamic repayment incentive dummy (Hidden Action Effect 2)	−0.019*	−0.000	−0.028**	0.004	−0.025**	−0.004	−0.080**	−0.000
	(0.010)	(0.017)	(0.011)	(0.021)	(0.012)	(0.020)	(0.032)	(0.057)
Dynamic repayment incentive size		−0.005		−0.009**		−0.006		−0.023*
		(0.004)		(0.004)		(0.005)		(0.013)
Offer rate (Hidden Information Effect)	0.005	0.004	0.002	0.002	0.007	0.007	0.015	0.015
	(0.003)	(0.003)	(0.003)	(0.004)	(0.005)	(0.005)	(0.011)	(0.012)
Observations	4348	4348	4348	4348	4348	4348	4348	4348
Adjusted R-squared	0.08	0.08	0.14	0.15	0.06	0.06	0.10	0.11
Probability(both dynamic incentive variables = 0)		0.06		0.00		0.06		0.01
Probability(all 3 or 4 interest rate variables = 0)	0.0004	0.0005	0.0003	0.0012	0.0006	0.0016	0.0000	0.0001

*significant at 10%; **significant at 5%; ***significant at 1%. Each column presents results from a single OLS model with the RHS variables shown and controls for the randomization condtions: observable risk, month of offer letter, and branch. Adding loan size and maturity as additional controls does not change the results. Robust standard errors in parentheses are corrected for clustering at the branch level. "Offer rate" and "Contract rate" are in monthly percentage point units (7.00% interest per month is coded as 7.00). "Dynamic repayment incentive" is an indicator variable equal to one if the contract interest rate is valid for one year (rather than just one loan) before reverting back to the normal (higher) interest rates. "Dynamic repayment incentive size" interacts the above indicator variable with the difference between the lender's normal rate for that individual's risk category and the experimentally assigned contract interest rate. A positive coefficient on the Offer Rate variable indicates hidden information, a positive coefficient on the Contract Rate or Dynamic Repayment Incentive variables indicates hidden action (moral hazard).

The dependent variable in columns (7) and (8) is a summary index of the three dependent variables used in columns (1)–(6). The summary index is the mean of the standardized value for each of the three measures of default.

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